

# Dexter Fernandes

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https://github.com/Dexter-Fernandes

## Data Scientist (Computer Vision)

Computer vision engineer, 3+ years building production video pipelines start to finish: training the models, then shipping them in containers to edge boxes that run 24/7 with nobody watching. I write real-time multi-stream detection and tracking in Python and C++17 on NVIDIA DeepStream and TensorRT. I hardened one deployed pipeline until live crashes dropped 67%, and cut false positives 33% on another. MSc Robotics, Bristol; LiDAR SLAM.

## WORK EXPERIENCE

### GrowthStage • Remote

05/2026 - Present

#### Recruiter

- Evaluated computer vision and SE candidates on their technical depth across SE practices, model deployment, and MLOps practices, maintaining direct exposure to current production engineering standards.

### Taco Bell • Bristol, England

09/2024 - Present

#### Service Champion / Team Trainer

### Esbaaar • Muscat, Oman

01/2023 - 08/2023

#### AI Engineer

- Owned a real-time NVIDIA DeepStream 6.x video inference pipeline running 24/7 on BP rigs, from model training and convergence debugging to production deployment, raising F1 15% via adversarial-condition training using PyTorch.
- Hardened the pipeline with Dockerised microservices, Kafka, structured logging, health checks and safe restarts, cutting live crashes 67% on unattended edge devices.
- Developed benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, EfficientDet) using PyTorch and torchvision, quantifying accuracy, latency and stability pre- and post-optimisation.
- Built OpenCV, Pillow and scikit-image preprocessing to detect mud and sand contamination on camera feeds, cutting false positives 13% in harsh industrial conditions.
- Applied Bayesian optimisation, ablation studies, hypothesis testing and confidence intervals in Jupyter, tracking experiments in Weights & Biases to support model selection, threshold tuning and failure analysis.
- Led peer code reviews in Git and introduced pytest and unittest coverage with linting into CI/CD validation workflows for edge inference modules, improving release consistency across hardware targets.

### LivNSense Technologies • Bengaluru, India

05/2020 - 12/2022

#### Lead Software Engineer - AI

- Led the development of a two-stage person detection and helmet-colour classification pipeline in TensorFlow and Keras, cutting false positives 33% for Austin Industries.
- Rebuilt the inference path in C++17 with OpenVINO on Intel VPU using object-oriented design and asynchronous multi-processing, raising throughput from 5 to 15 FPS within fixed memory and latency budgets.
- Compressed models 50% through quantisation-aware training, structured pruning and layer fusion in ONNX and OpenVINO while holding accuracy targets.
- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost-per-inference.
- Designed the PostgreSQL schema routing real-time incidents into downstream analytics, writing the joins and CTEs behind operational reporting.
- Automated labelling and dataset versioning, cutting manual labelling effort by 80% and reducing model rollout time from 72 hours to under 24.

## PROJECTS

### Presentation-Attack Detection under Generative AI

06/2026 - Present

- Building a face presentation-attack detection pipeline (photo, replay, and mask attacks), combining classical texture and frequency-domain features with a fine-tuned CNN, evaluated using APCER/BPCER/ACER/EER.
- Designing cross-dataset generalisation tests (training on CelebA-Spoof, evaluating on NUAA for print attacks; intra-dataset domain-shift protocol for replay/mask) alongside two augmentation strategies -- adversarial perturbation and procedurally-synthesised attack artefacts -- with Grad-CAM applied for failure-mode interpretability.
- Incorporating a subgroup-level fairness audit using a documented ITA skin-tone proxy, controlling for illumination and attack-medium confounds.
- Targeting ONNX/TensorRT edge deployment with latency profiling for real-time inference feasibility.

### DeepStream RTSP Pipeline

06/2026

- Built a multi-stream NVIDIA DeepStream 9.0 / GStreamer RTSP video analytics pipeline running YOLO detection with NvDCF tracking, metadata extraction, anonymisation and re-streamed output across 3 concurrent camera feeds.
- Optimised a YOLO26n TensorRT FP16 detector on 6GB Turing hardware using dynamic-batch engine conversion and p99 latency profiling, sustaining 29.7 FPS per stream live and 131.3 FPS unthrottled, validated FP16 against FP32 for accuracy.
- Implemented a C++/CUDA TensorRT plugin (IPluginV2DynamicExt) moving post-processing into the inference engine.
- Hardened with GPU smoke tests, model-promotion gates, structured JSON logging and per-camera health metrics covering liveness, dropped frames and reconnect events.

### Modern Edge CV Benchmarking

03/2026

- Benchmarked YOLOv8, YOLOv10, RT-DETR, EfficientViT, MobileSAM and Depth Anything v2 on edge hardware, comparing accuracy, latency and memory under matched conditions.
- Built a repeatable harness to hold input resolution, batch size and precision constant across architectures, isolating model choice from deployment configuration.

## EDUCATION

### Master of Science in Robotics - Merit

University of Bristol • Bristol, England

09/2023 - 09/2024

#### Dissertation: Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system using RTAB-Map on a Clearpath Husky, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.

### Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

- Built a foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

## SKILLS

- **Computer Vision & Deep Learning:** ByteTrack, Data augmentation, EfficientDet, HuggingFace, Image classification, Keras, NumPy, NvDCF, NvSort, Object detection, Object tracking, OpenCV, Pillow, PyTorch, RT-DETR, Scikit-image, Segmentation, TensorFlow, Torchvision, YOLO
- **Edge Inference and Optimisation:** custom TensorRT plugins, FP16/INT8 Quantisation, GStreamer, Intel Movidius Myriad VPU, NVIDIA DeepStream, NVIDIA Jetson AGX Xavier, ONNX, OpenVINO, TensorRT
- **LiDAR & Point Cloud:** Denoising, Eigen, g2o, GTSAM, ICP (Iterative Closest Point), LiDAR extrinsics calibration, LiDAR SLAM, LIO-SAM, Open3D, Point Cloud Library, Pose graph optimisation, ROS2, RTAB-Map, Scan matching
- **Programmng, Cloud & MLOps:** Apache Airflow, Apache Kafka, Bash, C++17, CI/CD, CUDA C++, Docker, DVC (Data Version Control), Git, Git, Jupyter, Jupyter, Linux, Microsoft Azure, MLflow, Multi-processing, Optuna, PIL (Pillow), pytest, Python, REST APIs, SQL (PostgreSQL), unittest, wandb
- **Annotation & Interpretability:** Confusion-matrix-driven failure clustering, CVAT, GRAD-CAM, TIDE (Toolbox for Identifying Detection Errors)